

Exposition of Music

VR Exhibition

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ABSTRACT

Nam June Paik's first solo exhibition in 1963 titled "Exposition of Music - Electronic Television" marked the beginning of his career, and he is often referred to as the father of video art. In this project, we attempted to virtually recreate some of his exhibits and exhibition spaces that are currently unavailable to elucidate their meaning. Among the exhibits, the three most interactive and meaningful works, namely Prepared Pianos (4 units), Electronic Television (13 units), and Random Access, were recreated in virtual reality (VR) space. To compensate for the limitations imposed by simple interactions between existing VR exhibitions of art works, this project was developed with the utmost focus on preserving the artistic value of the original work while maximizing the exhibition space reproduced in VR as well as audience immersion in the work.

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1 INTRODUCTION

Owing to the influence of the current COVID-19 pandemic, offline exhibitions are unable to cater to audiences, and museums and art centers are experiencing difficulties. Efforts are therefore underway to resolve difficulties with access to such exhibitions, especially since the audiences cannot personally visit these exhibits themselves. Recently, the Louvre Museum of France [Evrard and Krebs, 2018] and the New York Metropolitan Museum [Hill and Mystakidis, 2012] have constructed realistic content exhibitions that incorporate virtual reality (VR) technology to develop systems that allow audiences to view their collections and art centers without having to physically visit them. However, current VR-based exhibits have been created by photographing and implementing the existing spaces in 360° based on due diligence or by reproducing



Figure 1: Prepared Pianos

the existing works using 3D graphics technology; further, simple interactions with these works have been developed using simple gestures or buttons on the VR controllers. These VR exhibitions have the disadvantages of not being able to appropriately convey the artistic value of the original works as experienced in offline spaces.

Therefore, the current project was developed with specific emphasis on preserving the artistic value of the original work while maximizing both the offline exhibition space and audience immersion in the work. In other words, this work aims to maximize the perceptual as well as psychological immersive effects to users through a storytelling-based interactive exhibition using VR technology. In this project, the main focus was on reproducing "the sublimation of media technology as an art of media technology," which the pioneer of video art Nam June Paik intended to express.

2 DESIGN AND IMPLEMENTATION

In March 1963, Nam June Paik's first solo exhibition titled "Exposition of Music - Electronic Television" was held at the Gallery Parnas in Wuppertal, Germany. This event could also be considered as the beginning of the video art era. In reproducing this meaningful exhibition in the virtual space, we wanted to provide the audience with a heightened experience than reproducing the work of Nam June Paik alone. Although we testified to his work and space through several black and white photographs and documents that are available to this day, we also attempted to realize through VR the real intent that Paik wanted to convey to the audience through this exhibition. Hence, we allowed the audience to freely explore and touch the exhibition spaces and works as if they were real by developing with Unreal Engine and VR headsets like Oculus Quest and Vive.

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Figure 2: Electronic Televisions



Figure 3: Random Access

One part of the exhibition, as shown Figure 1 is named “Prepared Pianos”, included four mounted pianos, of which two had various objects suspended from, attached to, or inserted through the keyboard as well as strings and intertwined with wires. The visitors were freely allowed to play on these pianos and observe the changes, such as hearing unfamiliar sounds when pressing on the keys, movement of objects with devices, or even operating lights, sirens, ventilators, recorders, and radios around them. Thus, the audience could directly interact with the works that would have otherwise only been observed and heard through their eyes and ears, respectively, in the offline scenario, thereby experiencing and becoming a part of the work.

In a second part of the exhibition named “Electronic Televisions” as shown Figure 2, Paik presented 13 experimental televisions, among which “Cuba TV” allowed an image to vary depending on the frequency of the music flowing into a television connected to a recorder. In addition to watching and listening to the electromagnetic waves on the TV screen in the virtual space, the audience could manipulate the TV directly by turning the dial on the “One Point TV” using their hands.

The third and final reproduction of Paik’s first solo exhibition was “Random Access,” as shown Figure 3 which was a pioneering work on arbitrary sounds and connections as participatory works that allowed visitors to hear sounds from cassette tapes geometrically arranged on a wall using electricity or to randomly select and listen to records on bars. In addition, “Random Access” was the most central concept and important point of debate in Paik’s work as it included concepts regarding interaction, participation, coincidence, and non-determinism.

Through this VR project, we recreated the currently nonexistent works of an artist in the virtual space in the form of real-time interactions and virtually re-implemented the exhibition space from Wuppertal, which does not currently exist, so that users could experience the works beyond the constraints of time and space.

3 DISCUSSION

This project was a re-implementation of the first solo exhibition by Nam June Paik at the Gallery Parnas in Wuppertal, Germany, in 1963, using VR technology. In particular, with the implementation of the realistic interaction design, this work intended to increase the sense of existence and immersion by allowing the visitors to enjoy active VR experiences. This is considered a meaningful project as it demonstrates the synergistic convergence of art and technology via re-examination of the meaning of Paik’s exhibition, which has great significance in art history, and serves as an example of convergent art.

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